Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
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M1

Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials

1 Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials

Study focus: Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000038 · EV0000039

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module two on nanomaterials and this will be the last class of this module.

Key details

- And finally, we will look at some applications of nanomaterials.
- When we talk of mechanical properties, the most important thing we need to understand here that the strength of the nanomaterials is found to be greater than the bulk material.
- What does it mean?
- When we take a substance in the form of a bulk and it has a certain strength.
- But when you come to the nanomaterial range, it is found that the strength increases to a great extent.
- Bulk copper we know can be bent with certain amount of ease.
- We take these copper particles in the nano range, like when you get it to below 50 nanometers, it is found that they become super hard in nature.
- The same copper which in the form of bulk could be bent cannot.
- We bend now and it becomes super hard and it loses all its malleability and deductivity, which we find in the case of bulk copper.
- The reason is that when you take a particle in the nano range, the copper.

- in the nano range the number of defects becomes very less there is less probability of defects occurring now what do we mean by this.
- When we have large lattice lattices as in the case of bulk material, we know one layer above the other is getting formed when the metal is being when we have a metal.
- There are one after the other layers are getting formed.
- In these layers we will find that there could be imperfections like dislocations, kinks, and vacancies.
- And because of these imperfections or these vacancies and dislocations, etcetera, we find that the strength of the material.
- [ASR low confidence; check audio] material becomes less, but when we.
- Have particles in the nano range, we know that we are forming very small lattice which has a smaller cross section, and therefore the probability of finding these imperfections is very less, and therefore we find that when we go to this nano range of such a small particle.
- Of less than 15 nanometer, definitely the number of imperfections is less.
- The first property we need to discuss is the melting point.
- What would be phase transition temperature when you change from solid to liquid.
- That temperature is also reduced, or other phase transitions could also happen with two different types of solids.
- Also, the same solid existing in polymorphic forms that also would be a phase transition.
- Point also decreases.
- They are very characteristic.
- In fact, melting point, if you want to check the purity of a metal the fur or of any material, the first thing we do is check the melting point.
- And that would be characteristic of the material.
- And we know if the material if it is showing this melting point.
- Point which is listed, then we know that the material we have taken is pure.
- But the same material in the nano range now the melting point becomes less.
- What is the reason for that?
- It's very simple to understand.
- We know that.
- There are a lot of surface atoms in the nanomaterials because the particles are so small, most of the atoms are on the surface.

Example or application from the lecture

- Now let us take an example of copper.
- We have a very good lattice, and that results in making copper, for example, a super hard material.
- Let us talk of thermal properties.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 05:00–09:59

Obviously, you need to break the bonds.

Key details

- Now these surface atoms are held with less energy, so less energy will be required to break these bonds, and therefore we have lesser bonds, so the less energy.
- energy will be required to break these parts and hence we find that the melting point is going to decrease.
- So because of these large number of surface atoms, we find that the melting point of materials in the nano range becomes less.
- Another property which changes is the thermal conductivity.
- It is found that there is a decrease in thermal conductivity.
- It is due to the vibration of these lattices.
- obviously it will result in scattering which is known as phonon scattering and once this scattering happens that means it is going to result in a decrease in the thermal conductivity.
- We find that the magnetic properties of nanomaterials are distinctly different from that of the bulk materials.
- You take a material in the bulk form, and when you go to the nano form.
- Range of the same material, the magnetic properties are found to be very different.
- Iron cobalt and nickel.
- But the iron cobalt and nickel when they are taken in the nano range, we find that this ferromagnetism changes to superparamagnetism.
- in the nanometer range.
- The magnetism doesn't come to zero when we remove the magnetic field in the case of a ferromagnetic materials.
- But the same material now I don't when it is taken in the nano range.
- We find that they change into paramagnetism.
- They shall start showing paramagnetic behavior.
- The material gets aligned in a magnetic field but to a very small extent.
- But in the case of nanomaterials.
- The material which was ferromagnetic does become paramagnetic, that means it loses its magnetism to a great extent.
- but in the presence of the magnetic field we find that the magnetic susceptibility is very high.
- Because we know that the surface energy is very high.
- Because the surface energy is high, now these magnetic dipoles they can flip.
- These will flip in the in the presence of the magnetic field, they will they can flip at room temperature itself, they can flip directions and therefore they lose their ferromagnetic nature.
- But nevertheless, when there is a magnetic field, they will get.
- Magnetized like a paramagnetic material.
- But the magnetization is going to be very high, so that the magnetic susceptibility is going to be very high in this case.
- And when you remove the magnetic field, they come back to zero.

Example or application from the lecture

- Let us talk of magnetic properties.

- Let us take up the example of.
- Let us first understand what exactly is a ferromagnetic material from this graph and what is a paramagnetic material so that we can understand what is that change.

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Core explanation · 10:00–14:59

behavior so from ferromagnetic it changed to superparamagnetic behavior why the reason is the high surface energy so the domains at room temperature itself flip directions and therefore they lose their alignment with the ma alignment and therefore when the magnetic field is applied they do get some small amount of alignment like paramagnetic behavior but the susceptibility is now very high.

Key details

- They get magnetized to a great extent and that is called superparamagnetic behavior but the g the interesting part is that when the magic field is removed they don't behave like ferromagnetic beh material which can retain they have memory but here it becomes zero.
- They are non-magnetic, but then in the nano size range they start behaving like magnetic particles.
- We know that nanomaterials have a range of applications in many fields.
- Then energy storage, catalysis, then consumer electronics, environmental and cosmetics.
- And then we can also have injectable nanopods which can serve as detectors, informers of early disease and perhaps repair metabolic defects.
- And apart from that we know that the there's.
- Silver nanoparticles are known for their antibacterial activities.
- There are several more applications in medicine.
- The region of energy storage.
- The electrodes which are used in fuel cells.
- The electrodes need to have very high.
- surface area because that is where the actual reaction takes place.
- The electrode is in contact with the fuel or the oxidant, it is also in contact with the catalyst.
- The catalyst being is being impregnated onto the electrode and it's also in contact with the electrolyte.
- Therefore a high surface in the or area of the electrode which would help in making the fuel cells very efficient.
- Then coming to catalysis.
- Now nano crystalline materials again they have a very large surface area because they are in they have a very small size and most of the atoms we find on the surface and these atoms will also be very active so with regard to their physical, chemical and mechanical properties, so we know that they can serve as very good catalysts.
- If we use the material in as a nano particle.
- In the nanoparticle range, we find that they will serve as a very good catalyst.

- And in fuel cells, again, oxygen reduction reaction is a very slow reaction.

Example or application from the lecture

- Now another property is the another example is where we have bulk, gold and platinum.
- For example, when we have these cancer cells and some drugs are given, we know that the they don't not only affect the cancer cells, but the cell around are also getting affected.
- For example, we know gold and silver nanoparticles are being used as catalysts in several experiments.

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Core explanation · 15:00–19:59

while make for very good catalyst and also as a catalyst for carbon monoxide oxidation.

Key details

- They are composed of materials which are known as phosphors which glow when they are struck with a stream of electrons from the within the catchment rate two.
- Size or the phosphorus.
- So obviously, if we take nanophosphors, then the size is sufficiently reduced, and therefore we find the number of pixels is more, and therefore we will find that the resolution is going to be much higher.
- materials like zinc selenite, then cadmium sulphide, zinc sulfide and lead sul tellurite.
- And nanophosphors are also being employed in white light emitting diodes.
- Nanomaterials can be very useful and are being employed in catalytic converters.
- Now because they have high chemical activity and due to the high surface area, therefore, nanomaterials are being used as catalysts to remove the toxicity.
- And the noxious gases such as the nitrogen oxide and carbon monoxide in automobiles as well as for power generation equipment.
- no, earlier when sun blocks used to be used they would be white in colour.
- But now because.
- Of the use of these nanomaterials, now these same zinc oxide and titanium oxide, they are taken in the nanoparticle range, and it is found that they now because of the nanoparticle range they become transparent.
- out you don't get that whitish color on the face and they are now transparent.
- These are just some of the applications which we have listed over here but yes they have a several.
- Applications in the electronics industry and or many applications are found for nanomaterials.
- So in the next class we are going to start with unit two where we will study phase equilibrium.

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EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9970; only the JSON segmentation supplies the displayed timestamps.
Thermal properties (melting point), electrical properties, and magnetic properties (superparamagnetism), Graphene - structure and synthesis; applications of nanomaterials transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/041_Thermal_properties_melting_point_electrical_properties_and_magnetic_properties_superparamagnetism_Graphene_-_s__3ce99088727d358f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00012	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

C Revision and retrieval practice